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1 **Can our phones keep us safe? A content analysis of smartphone applications**
2 **to prevent mobile phone distracted driving**

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27 **Can our phones keep us safe? A content analysis of smartphone**

28 **applications to prevent mobile phone distracted driving**

29 Mobile phone use while driving is a pervasive problem that continues to increase,
30 notwithstanding the large crash risk this behaviour constitutes. A number of phone
31 applications have been developed with the intention of utilising the technology to prevent
32 dangerous phone behaviours while driving. Despite the potential these applications have
33 in preventing crashes associated with distracted driving, research is yet to explore these
34 emergent applications. Therefore, this study provided a review of the current smartphone
35 applications developed to prevent distracted driving. A content analysis was conducted
36 to identify the smartphone applications targeted at stopping, preventing or reducing
37 phone use behaviour while driving. Their functionality was determined based on the
38 ecosystem of smartphone applications: application-mobile phone interaction,
39 application-driver interaction, and application-context interaction. A total of 29 relevant
40 applications in English language were identified. Most of these applications focused on
41 blocking specific phone functions (e.g. texting or calling) while allowing more desirable
42 driving phone functions to be accessed (e.g. music applications and GPS functions). The
43 specific functions which are blocked or allowed varied greatly between applications. Out
44 of the different application interactions, the function which sends an automatic text
45 message to a contact who texts the driver (associated with external communicator
46 interactions) was the most common feature. A major limitation of the applications was
47 their reliance on blocking specific phone functions as opposed to managing workload
48 while driving or simplifying specific phone tasks to be more compatible with driving.
49 Simply blocking phone functions may not be attractive to drivers who view their phone
50 as a necessity. As such, these drivers are unlikely to use these voluntary applications at
51 all while driving. Smartphone applications designed to prevent phone use while driving
52 show potential for playing a large role in a systemic intervention to prevent mobile phone
53 distracted driving, yet there is a substantial need for further development of these
54 applications.

56 *Keywords:* texting; Human-Computer Interaction; cell phone; distraction; dual-task;
57 inattention; apple

58 *Highlights:*

- 59 • 29 smartphone applications which prevent distracted driving were identified
- 60 • These applications focus on blocking phone functions
- 61 • Ecosystem of smartphone applications was used to examine their functionality
- 62 • The applications do not focus on simplifying phone tasks while driving

63

1. Introduction

Mobile phone distracted driving is a type of inattention that occurs when drivers divert (or share) their attention away from the driving task to focus on a mobile phone activity. Mobile phone use while driving is a global concern that is estimated to increase crash risk. Based on crash reports in the U.S., it is believed that distracted driving (including mobile phone use) was involved in nearly 10% of all traffic fatalities in 2014 (NHTSA, 2016). In addition, a study of police crash reports found that mobile phone distraction resulted in 18% of fatal crashes and 5% of injury crashes (Overton et al., 2014). Asbridge et al. (2013) reported that the odds of a culpable crash increase by 70% when the driver is using a mobile phone.

On-road studies are also utilised to investigate changes in crash risk as a consequence of mobile phone use while driving. The most recent and largest naturalistic study in the U.S., Second Strategic Highway Research Program Naturalistic Driving Study (SHRP 2 NDS), gathered information from more than 3,500 drivers across a 3-year period (Dingus et al., 2016). The SHRP 2 NDS has reported that browsing, handheld dialling, and handheld text interactions with a mobile phone significantly increase the odds of crash risk by a factor of 2.7-12.2 (Dingus et al., 2016). On the other hand, cognitive-manual (e.g. handheld mobile phone conversations) and cognitive (e.g. hands-free mobile phone conversations) mobile phone interactions were not found to be associated with increased crash risk after controlling for “additional task” selection bias and “driver behaviour error” confounding bias (Young, 2017). These results have been further documented in both a recent meta-analysis (Simmons et al. 2016) and recent systematic review (Oviedo-Trespalacios et al. 2016) of on-road and naturalistic studies. The threat that mobile phone use while driving poses, particularly visual-manual mobile phone use, makes this one of the most important issues in road safety worldwide.

Little success has been reported worldwide in stopping mobile phone use while driving. Highly-developed road safety systems in countries such as Australia, the U.K, Spain, and Germany, with less than 6 road fatalities per 100,000 inhabitants per year (World Health Organization, 2015), have been unable to fully prevent distracted driving. For instance, a cross-sectional study in Australia showed that one in every two drivers ($n = 443$) engaged in an illegal task such as handheld conversations or texting/browsing while driving in a typical day (Oviedo-Trespalacios et al., 2017). In Spain, a study using roadside interviews ($n = 426$) found that 32.2% of drivers reported having handheld conversations and 43.7% reported sending text messages while driving (Prat et al., 2017). These studies report mobile phone use sometime during a trip or during a day, whereas observational studies indicate how many drivers are using mobile phones at a given time and location. In the U.K, an observational study of 10,984 drivers at random locations observed that nearly 2.7% of drivers were talking on a handheld or hands-free mobile phone (Sullman et al., 2014). Likewise, in Germany, an observational study (including 11,837 observations) detected a prevalence of 4.5% for visual-manual interactions such as texting or browsing while driving (Vollrath et al., 2016). It is important to note that many of these countries have legislation that prohibit handheld mobile phone use while driving, however, legislation has had a minimal effect due to the difficulties in enforcing the rule (Nevin et al., 2016) and enforcement-avoidance strategies used by drivers (Oviedo-Trespalacios, 2018). Distracted driving affects population health by increasing crash risk, and research exploring potential countermeasures is still much needed.

An emergent opportunity for distracted driving prevention is presented by smartphone applications designed to prevent mobile phone use while driving. The common rationale across these smartphone applications is that the mobile phones could prevent interactions while the vehicle is moving (Siuhi and Mwakalonge, 2016). A common example is the “Do Not Disturb While Driving” application developed by Apple (now preinstalled in iPhones worldwide) with the promise of helping drivers keep their eyes on the road by disabling phone functions such as texting once the iPhone senses vehicle motion (APPLE Inc., 2017). Reducing off-road glances while in control of a vehicle is very positive in terms of safety, given that sharing visual attention between mobile phone tasks and driving has been associated with crashes and impaired vehicle control (NHTSA, 2016, Simmons et al., 2016, Oviedo-Trespalacios et al., 2016, Oviedo-Trespalacios et al., 2018a). However, there is little evidence regarding the effectiveness of voluntarily initiated smartphone applications in preventing mobile phone related road crashes. A plethora of issues remain to be assessed concerning the capability of these smartphone applications to fully stop risky mobile phone use while driving and the intentions/acceptability of drivers regarding their use. Firstly, mobile phone use while driving is a complex activity (Oviedo-Trespalacios et al., 2017, Oviedo-Trespalacios et al., 2016), as drivers can engage in a wide range of activities that can increase crash risk (e.g., texting or browsing) as well as support driving (e.g. GPS or weather alerts). Secondly, research has suggested that some drivers develop maladaptive attachment to their mobile phones (Hong et al., 2012, Weller et al., 2013), which would reduce their willingness to activate these smartphone applications. Therefore, it is necessary to gain a better understanding of the potential safety benefits of these smartphone applications as well as the features that could influence their use among drivers.

This paper aims to review current smartphone applications designed to prevent distracted driving. A methodology was designed to investigate two main topics: (i) the safety implications of the features currently available in smartphone applications; (ii) the potential effectiveness of the smartphone applications to reduce mobile phone distraction. Finally, research gaps and future research were identified to support the implementation of smartphone applications as a tool to stop mobile phone distracted driving.

2. Methods

A content analysis of web pages promoting smartphone applications designed to prevent mobile phone use while driving was conducted. Content analysis is a methodology for making inferences from qualitative data by systematically characterizing its themes and has been frequently used in the analysis of websites, e.g. retirement village websites (Hu et al., 2017), cosmetic and anti-aging products and treatments websites (Rachul et al., 2015), and fast-food websites (Wellard et al., 2017).

2.1. Search Strategy

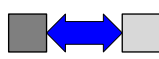
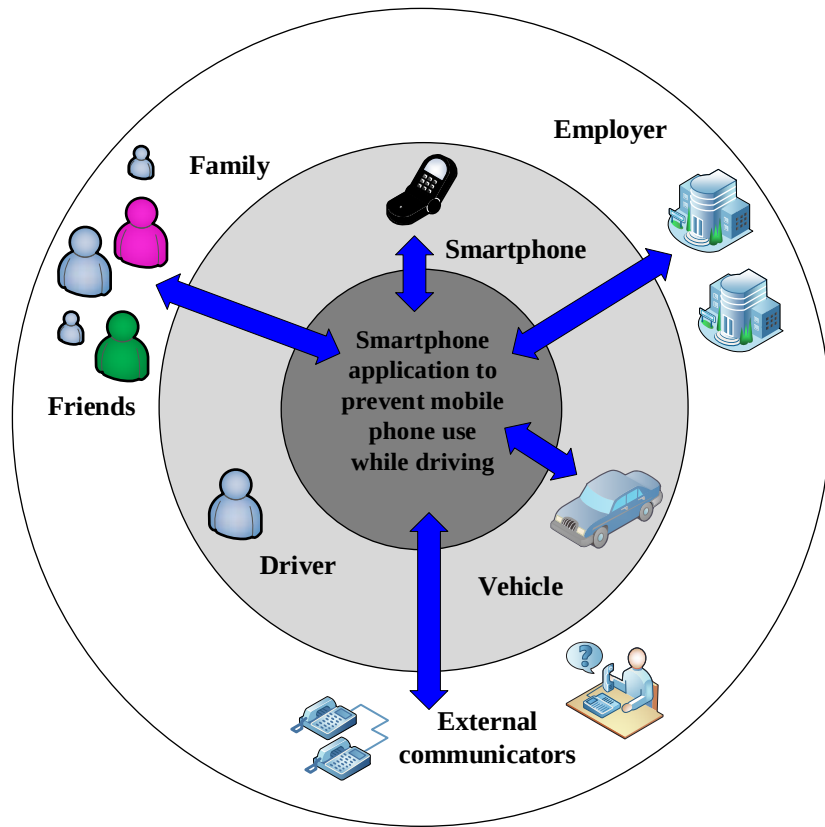
To identify the smartphone applications to be reviewed, a search on the Google search engine was conducted using the keywords “stop” AND “distracted” AND “driving” AND “app”. All the web pages reviewed were in English. Each web page was reviewed to identify potential smartphone applications including pages describing a specific application or listing a group of

applications. Every smartphone application mentioned in the aforementioned web pages was searched consecutively. Additional searches were conducted in the Google Play store for Android (85% of smartphone market) and the Apple iTunes store for iOS (14.7% of smartphone market) using the keywords listed above and also reviewing other suggested smartphone applications to identify the available applications in the market (see <https://www.idc.com/promo/smartphone-market-share/os>). As exclusion criteria, only smartphone applications targeted at stopping, preventing, or reducing mobile phone use while driving were considered and only web pages in English were used. The searches were conducted by the first author and the eligibility of the unique applications was decided through discussion among all the co-authors. A total of 36 unique applications were identified, of which 32 were eligible. Finally, three applications were discontinued during the study and therefore removed (see appendix).

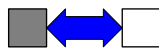
2.1. Information processing

To investigate the functionality of smartphone applications, a coding sheet was designed based on the ecosystem of smartphone applications to prevent mobile phone use while driving (see Figure 1). The coding sheet was developed in an iterative process. First, one author randomly selected 15 applications and initially coded the included definitions to develop a preliminary coding sheet. The preliminary coding sheet was based on the view of mobile phone distracted driving as a human-machine system proposed by Oviedo-Trespalacios et al. (2016), which suggests two levels of interactions: (i) smartphone application to prevent mobile phone use while driving and the human-machine system components (i.e., driver, car, and mobile phone) and (ii) smartphone application to prevent mobile phone use while driving and the environment. The coding system was revised through discussion among the authors. Next, all applications were coded following a two-step analysis where two researchers independently coded each application and continuously extending the coding sheet if new codes emerged during the analysis of new applications. A third reviewer was consulted to resolve any disagreement in extracted information such as inconsistency in characteristics of smartphone applications or interaction level (e.g., missing information). Following the recommendations of Hsieh and Shannon (2005), the codes were discussed within the research team and grouped into meaningful clusters. The searches and data extraction were conducted in December 2017. In order to provide updated information regarding the applications in terms of number version and download, two researchers updated this information on an ongoing basis. The last update was conducted in April 2018.

As seen in Table 1, five interactions were identified between the smart-phone applications and the ecosystem: application-mobile phone interaction, application-driver interaction, and application-context interaction, where context is divided into three areas – external communicators trying to contact the driver, external users (i.e., individuals' or organisations' interaction with the application)) and the vehicle.



Interaction level: smartphone application to prevent mobile phone use while driving and the human-machine system components (i.e., driver, car, and mobile phone)



Interaction level: smartphone application to prevent mobile phone use while driving and the environment

Figure 1. Ecosystem of smartphone applications to prevent mobile phone use while driving

212 **Table 1.** Coding Scheme

Interactions	Definition	Examples
Interaction level: smartphone application to prevent mobile phone use while driving and the human-machine system components (i.e., driver, car, and mobile phone)		
Smartphone application-mobile phone interactions	Interactions between the smartphone application to reduce distracted driving and the functioning of the mobile phone	• Enable calling function through any interface hands-free, handheld, speaker, etc. • Enable texting and other visual-manual interactions • Enable GPS and navigation applications • Enable music • Record mobile phone use
Smartphone application-driver interactions	Interactions between the smartphone application to reduce distracted driving and the driver	• Smartphone application reads text messages aloud • Driver can respond to text by voice with support of the smartphone application
Smartphone application-vehicle interactions	Interactions between the smartphone application and the vehicle	• Smartphone application automatically detects that the vehicle is moving • Smartphone application integrates with the vehicle through Bluetooth or hardware
Interaction level: smartphone application to prevent mobile phone use while driving and the environment		
Smartphone application-external communicator interactions	Interactions between the smartphone application and external communicators intending to reach the driver	• Smartphone application automatically responds to texts • Smartphone application automatically responds to calls
Smartphone application-external user interactions	Interactions between the smartphone application and external users (e.g., family or employer) seeking information	• Smartphone application shares recorded information such as mobile use while driving, e.g. how many text messages were sent while the application was active.

214 3. Results

215 A total of 29 applications advertised to stop/prevent/reduce distracted driving were identified.
 216 The applications and their main characteristics are presented in Table 2. The most common
 217 Android application was Android Auto. Given the “Do Not Disturb While Driving” function
 218 is pre-installed in all iOS 11 devices, this was the most common application for iOS phones.
 219 AT&T DriveMode was the most common application that was available for both Android and
 220 iOS devices. Almost all applications ($n = 28$) were available in Android phone systems.
 221 Meanwhile, 11 applications were available for iOS phone systems and only 2 applications were
 222 available on Blackberry phone systems. A total of 15 applications were free to use, 4
 223 applications needed to be purchased in order to be used and 10 applications included a mix of
 224 free use and the requirement to provide payment for use; this mostly involved either a free
 225 demonstration or the application was free to download but included in-application purchases.

227 **Table 2.** Smartphone applications for reducing mobile phone use while driving identified during the on-line
228 searches

Smartphone applications to reduce mobile phone use while driving	Version	Number of Installations	Access (free, pay, mix)	Platform		
				iOS	Android	Blackberry
Android Auto	3.1	10,000,000 – 50,000,000	Free		x	
AT&T DriveMode	3.3	10,000,000 - 50,000,000	Free	x	x	
BAZZ Text While Driving Safe	2.45	50,000 - 100,000	Free		x	
Car Mode	7.0	50,000 – 100,000	Free		x	
Cellcontrol	1.78	10,000 - 50,000	Mix (free demo)	x	x	
cellMUTE+ Drive Mode & More	2.62	10,000 - 50,000	Mix (in app purchases)		x	
Do Not Disturb While Driving	N/A	Pre-installed IOS11	Free	x		
Down For the Count	2.1.0	Not stated	Free		x	
Drive Now Text Later	3.1	1,000 - 5,000	Mix (in app purchases)		x	
Drive Safe	3	10,000 - 50,000	Mix (in app purchases)		x	
DriveAlertNow	N/A	10-50	Pay	x	x	
DriveSafe.ly	3.8	1,000,000 - 5,000,000	Mix		x	x
eBrake	N/A	N/A	Mix	x	x	
Fleetmode	5	100-500	Mix (free demo)		x	
Lifesaver	7.0.4	5,000 - 10,000	Mix (free demo)	x	x	
Live-n-Drive	7	500 - 1,000	Free		x	
MessageLOUD	5.98	50,000 – 100,000	Pay	x	x	
Mojo: Rewards for Safe Driving	1.4.0.804	1000 - 5000	Free	x	x	
No Texting While Driving App	1.3	5,000 - 10,000	Free		x	
OneTap—Block Phone Alerts	1.5.1	10,000 - 50,000	Free		x	
ProtextMe Safe Driving FREE	4.1	10,000 - 50,000	Free		x	
PULL OVER TO TEXT	4.4	100 - 500	Pay		x	
Safe Driving + Auto SMS	6.0	100 - 500	Pay		x	
Safe Driving App: Drivemode	7.1.1	1,000,000 - 5,000,000	Mix (in app purchases)		x	
Safely Go	1.2	50,000 - 100,000	Free		x	
Text Ninja	3.21	100 - 500	Free	x	x	
TextDrive	2.5.7	10,000 - 50,000	Mix (in app purchases)		x	
TextLimit	2.0	500 - 1000	Free	x	x	x
TrueMotion Family Safe Driving	1.10.2	10,000 - 50,000	Free	x	x	

229
230 Table 3 identifies the overall smartphone functions the applications allow and do not
231 allow, as well as whether the applications record phone use while driving. A total of 26
232 applications allow calls and 10 applications allow texting and other visual-manual tasks.

However, many of the applications differ in the extent to which they allow calls and visual-manual tasks, 22 of the applications allow navigation, 20 applications allow the user to listen to music and 12 applications record the driver's phone use.

Table 3. Characteristics of smartphone applications for reducing mobile phone use while driving as reflected in the application eco-system: smartphone application-mobile phone interactions

Smartphone applications to reduce mobile phone use while driving	Smartphone application-mobile phone interactions				
	<i>Calls</i>	<i>Texting and other visual-manual tasks</i>	<i>Maps/Navigation</i>	<i>Music</i>	<i>Record Phone use</i>
Android Auto	Yes (make calls with Google Assistant and answer calls with a tap)	Yes (via google assistant)	Yes	Yes	No
AT&T DriveMode	Yes (will silence call but will not block call)	Silence notifications	Yes	Yes	No
BAZZ Text While Driving Safe	Yes	No	No	No	Yes
Car Mode	Yes	Yes	Yes	Yes	No
Cellcontrol	Yes (Can add white list numbers)	No (however, this is customisable)	Yes	Yes	Yes
cellMUTE+ Drive Mode & More	Using hands-free	No	Yes	No	No
Do Not Disturb While Driving	Yes (can be restricted to few contacts)	Silence notifications (if desired by the sender)	Yes	Yes	No
Down For The Count	Yes	Yes	Yes	Yes	Yes
Drive Now Text Later	Using hands-free	No	No	No	No
Drive Safe	Yes (after 3 incoming calls)	Silence notifications	Yes	Yes	No
DriveAlertNow	Yes (restricted to few contacts)	No	Yes	Yes	Yes
DriveSafe.ly	Yes	No	Not stated (unknown if other apps work at the same time)	Not stated	No
eBrake	Yes	Silence notifications	Yes	Yes	No
Fleetmode	No	No	Yes	No	Yes
Lifesaver	Using hands-free	No	Yes	Yes	Yes
Live-n-Drive	Silent/vibration (call will be loud after 2 calls in pre-determined time)	Silence notifications	No	No	No
MessageLOUD	Yes (automatically answer calls and respond to calls with eyes-free screen tap)	No	Yes	Yes	No
Mojo: Rewards for Safe Driving	Yes	Yes	Yes	Yes	Yes
No Texting While Driving App	No	No	Yes	Yes	No
OneTap—Block Phone Alerts	Yes (asking the driver to pull over)	No	Not stated (unknown if other apps work at the same time)	Not stated	Yes
ProtextMe Safe Driving FREE	Yes	Silence notifications (can create exceptions)	N/A	N/A	Yes
PULL OVER TO TEXT	Yes	No	Yes	Yes	No
Safe Driving + Auto SMS	Yes (emergency list contacts only)	No	Yes	Yes	No
Safe Driving App: Drivemode	Yes (option to ignore calls in “do not disturb” mode)	Yes (limited – only using voice commands)	Yes (allows use of 3 chosen apps)	Yes (allows use of 3	Yes

				chosen apps)	
Safely Go	Yes (only from 3 chosen contacts)	Yes (only from 3 chosen contacts)	Yes	Yes	No
TextLimit	No	No	No	No	No
Text Ninja	Yes	Silence notifications	Yes	Yes	Yes
TextDrive	Yes (will only receive calls from contact that calls more than once)	Yes	Yes	Yes	No
TrueMotion Family Safe Driving	Yes	Yes	Yes	Yes	Yes

Table 4 outlines the possible interactions of each application, including driver interactions, external communicator interactions, external user interactions and vehicle interactions. The automatic response to text message component of external communicator interactions was the most common feature among the different applications, closely followed by the detecting moving vehicle factor linked to vehicle interactions. The least common application function was the ability to respond to texts by voice, which is linked to the driver's interaction with the application.

246
247

Table 4. Characteristics of smart phone applications for reducing mobile phone use while driving as reflected in the application eco-system: smartphone application-driver interactions, smartphone application-external communicator interactions, smartphone application-external user interactions, and smartphone application-vehicle interactions

Smartphone application to reduce mobile phone use while driving	Smartphone application-driver interactions		Smartphone application-external communicator interactions		Smartphone application-external user interactions	Smartphone application-vehicle interactions	
	<i>Read text aloud</i>	<i>Respond to text by voice</i>	<i>Respond to text automatically</i>	<i>Respond to calls automatically</i>	<i>Share information</i>	<i>Detect moving vehicle</i>	<i>Integrate with vehicle</i>
Android Auto	Yes	Yes	Yes	No	No	Yes	Yes (Bluetooth)
AT&T DriveMode	No	No	Yes	No	Yes	Yes	Yes (Bluetooth)
BAZZ Text While Driving Safe	Yes	Yes	No	No	No	Yes	Yes (Bluetooth)
Car Mode	Yes	Yes	No	No	No	Yes	Yes (Bluetooth)
Cellcontrol	No	No	No	No	Yes	Yes	Yes (hardware)
cellMUTE+ Drive Mode	No	No	No	No	No	Yes	No
Do Not Disturb While Driving	Yes	Yes	Yes	No	No	Yes	Yes (Bluetooth)
Down For The Count	No	No	Yes	No	No	No	No
Drive Now Text Later	No	No	Yes	Yes	No	Yes	No
Drive Safe	No	No	Yes	Yes	No	Yes	No
DriveAlertNow	No	No	Yes	Yes (transfer voicemail)	Yes (if device is overridden and statistics)	Yes	Yes (Bluetooth)
DriveSafe.ly	Yes	Yes	Yes	No	No	No	Yes (Bluetooth)
eBrake	No	No	Yes	Yes	No	Yes	Yes (Bluetooth)
Fleetmode	No	No	Yes	Yes (transfer voicemail)	Yes	Yes	Yes (hardware)
Lifesaver	No	No	Yes	No	Yes	Yes	No
Live-n-Drive	No	No	Yes	Yes	No	Yes	Yes (Bluetooth)

MessageLOUD	Yes	No	No	Yes	Yes	Yes (automatically answer calls or auto-respond to calls)	Yes (Bluetooth)
Mojo: Rewards for Safe Driving	No	No	Yes	Yes	No	No	No
No Texting While Driving App	No	No	Yes	Yes	No	Yes	No
OneTap—Block Phone Alerts	No	No	Yes	Yes	Yes	Yes	No
ProtextMe Safe Driving FREE	Yes	No	Yes	No	Yes (The app will send a message to the management team if the employee texts while driving)	Yes	Yes (Bluetooth)
PULL OVER TO TEXT	No	No	Yes	No	No	Yes	No
Safe Driving + Auto SMS	Yes	Yes (pre-recorded voice message)	Yes	Yes (missing calls)	No	Yes	No
Safe Driving App: Drivemode	No	Yes	Yes	No	Yes	No	Yes (Bluetooth)
Safely Go	No	No	No	No (needs other application to do this)	Yes	Yes (Transfer to voicemail)	Yes (Bluetooth)
Text Ninja	No	No	Yes	Yes	No	No	Yes (Bluetooth and hardware)
TextDrive	Yes	Yes	No	No	Yes	Yes	No
TextLimit	No	No	No	Yes	No	No	No
TrueMotion Family Safe Driving	No	No	Yes	Yes	No	No	No

249 A summary of features from the smartphone applications designed to prevent
 250 distracted driving, identified in the content analysis, is presented in Table 5. Most of the
 251 applications (89.7%) offer solutions to manage mobile phone calls while driving, while
 252 nearly half (51.7%) target texting and other visual-manual task management while
 253 driving. Nearly a third of the applications offer solutions to substitute visual-manual
 254 interactions with voice/audio interactions. A total of 55.2% of the applications can be
 255 integrated with the vehicle.

256 **Table 5.** Summary of content analysis

Interactions	Features	Frequency	%
Smartphone application-mobile phone interactions	Calls	26	89.7%
	Texting and other visual-manual tasks	15	51.7%
	Maps/Navigation	22	75.9%
	Music	20	69.00%
	Record Phone use	12	41.4%
Smartphone application-driver interactions	Read text aloud	9	31.0%
	Respond to text by voice	8	27.6%
Smartphone application-external communicator interactions	Respond to text automatically	21	72.4%
	Respond to calls automatically	14	48.3%
Smartphone application-external user interactions	Share information	11	37.9%
Smartphone application-vehicle interactions	Detect moving vehicle	22	75.9%
	Integrate with vehicle	16	55.2%

257

258 4. Discussion

259 There are a great number of smartphone applications that are promoted for the
 260 prevention of mobile phone distracted driving. The content analysis confirmed that most
 261 of the smartphone applications available today are focused on blocking functions such as
 262 texting, browsing or handheld conversations. This focus on blocking visual-manual tasks
 263 is encouraging, considering that recent studies have consistently identified the risks
 264 associated with drivers taking their eyes off the road to use a mobile phone (NHTSA,
 265 2016, Simmons et al., 2016, Oviedo-Trespalacios et al., 2016). Research has suggested
 266 that some drivers using smartphone applications such as *DriveSafe.ly* might reduce their
 267 exposure to texting (McGinn, 2014). However, a number of applications to prevent
 268 mobile phone use while driving still allow texting applications, some attached to
 269 applications that are not primarily text based, e.g., Facebook Messenger, Viber,
 270 WhatsApp, WeChat, etc.

271 Another issue is that mobile phone devices have significantly changed over the
 272 years to include technologies that allow engagement in a wide range of activities such as
 273 navigating social media (e.g., Instagram, tinder, Facebook, twitter), GPS, listening to
 274 music, and internet gaming, all on one device, which has increased mobile phone use in
 275 our lives (Ayar et al., 2017). Findings from this research show that applications to prevent
 276 mobile phone use while driving might not fully prevent visual-manual interactions such
 277 as in-car streaming music interfaces or GPS devices, which is not always compatible with

278 driving. Recent studies found that young drivers have a strong intention to use music and
279 GPS phone functions while driving (George et al., 2018, Truelove et al., 2019). There is
280 a need to define strategies that support the use of these functions in a risk-minimising way
281 while driving.

282 Functions targeting involuntary distraction remain relatively underdeveloped in
283 smartphone applications to prevent distracted driving. Involuntary distraction involves
284 sudden non-driving information or stimuli originated from the mobile phone that captures
285 the driver's attention, i.e., notifications of incoming text messages, email or calls.
286 Incoming calls and, less frequently, notifications are permitted in most of the smartphone
287 applications identified in this review. A sudden incoming call/notification represents a
288 greater risk for traffic safety compared to a self-initiated interaction. Externally triggered
289 tasks do not consider current driving demands and prevent drivers from managing
290 workload (Oviedo-Trespalacios et al., 2017, Oviedo-Trespalacios et al., 2018a, Oviedo-
291 Trespalacios et al., 2019). Results from recent naturalistic studies confirm that activities
292 such as reaching for a mobile phone can increase crash risk (odds ratio of 4.8) (Dingus et
293 al., 2016). Findings from this study show that not many applications manage incoming
294 calls. Some smartphone applications restrict incoming calls to the use of hands-free
295 interfaces (e.g., cellMUTE+ Drive Mode), which avoid visual-manual demands. On the
296 other hand, some applications restrict incoming calls to a limited selection of contacts
297 (e.g., DriveAlertNow), which could minimise exposure. There are no applications that
298 help drivers to safely manage an incoming call based on the level of driving demand being
299 experienced.

300 One of the most important findings of this investigation is that the smartphone
301 applications developed for preventing mobile distracted driving lack variety. The
302 widespread task-blocking approach is not flexible enough to safely manage instrumental
303 mobile phone interactions with Maps or Music applications. In addition, blocking tasks
304 might not benefit some groups of the population that have developed strong habits of
305 mobile phone use and, therefore, might not accept this technology (Bianchi and Phillips,
306 2005). Studies conducted in Australia and the U.S. have demonstrated that "mobile phone
307 blockers" have low acceptability among drivers, which reduces technology adoption
308 (Delgado et al., 2018, Ponte et al., 2016). An option to overcome this limitation is to shift
309 the mobile phone task-blocking paradigm towards the design of strategies seeking better
310 compatibility between mobile phone tasks and driving. Authorities such as the NHTSA
311 (2016) have reported that the development of technologies dedicated to reducing driver
312 workload (i.e., visual, manual, and cognitive demands) associated with performing
313 secondary tasks is a worthwhile approach to increase safety.

314 Smartphone applications that facilitate safer mobile phone use while driving are
315 in the early stages of design and development. However, findings from the content
316 analysis confirm that some smartphone applications have started to incorporate safe
317 mobile phone-driving integration principles into their designs. Two main approaches
318 have been considered: (i) changing the nature of the interaction between the driver and
319 the phone; and (ii) simplifying smartphone functionality. The nature of the interaction
320 can be changed by the use of voice commands instead of the visual-manual interface, and
321 is emerging in smartphone applications such as *Do Not Disturb While Driving* and

322 *DriveSafe.ly*, which are capable of reading text aloud and responding to texts by voice.
323 As explained earlier, visual-manual interactions are a major risk for mobile phone
324 distracted drivers (Dingus et al., 2016). Although some studies have concluded that the
325 impact of phone conversations is negligible (Young, 2017, Simmons et al., 2016, Oviedo-
326 Trespalcacios et al., 2016), there are concerns that the cognitive demands of conversations
327 might be risky under certain circumstances such as difficult conversations (Lansdown
328 and Stephens, 2013). Nonetheless, exchanging visual demands for audio/cognitive
329 demands is a viable risk-minimising strategy. Unfortunately, the technology is not always
330 reliable and drivers are not likely to use voice commands to send text message unless they
331 are experiencing adverse driving conditions (Tchankue et al., 2012). Low reliability and
332 accuracy of the technology is likely to undermine widespread implementation of these
333 smartphone applications. However, mobile phones are evolving to have increased
334 capabilities to monitor driving behaviour (Kanarachos et al., 2018), which will certainly
335 benefit future developments in this area.

336 On the other hand, simplification of mobile phone functions is a new concept
337 which enables drivers to engage in a wide range of tasks including texting. The
338 smartphone application *Safe Driving App: Drivemode* is the only application identified
339 in this study based on this concept. *Drivemode* allows a wide range of activities from
340 driving related applications such as GPS to social media interactions and texting. In
341 addition, *Drivemode* uses simplified interfaces to integrate the mobile tasks into driving
342 with the aim of reducing visual and motor demands during driver-phone interactions, e.g.,
343 providing shortcuts to relevant actions that the driver performs with his/her mobile phone.
344 In addition, *Drivemode* has a number of voice-based commands to substitute driver
345 demands. Although applications supporting safe integration between mobile phone tasks
346 and driving are becoming more widespread, caution is still required in recommending
347 their implementation. At present, there is no empirical evidence that these workload
348 manager applications are able to reduce crash risk. In addition, concerns remains that,
349 under some road traffic conditions, drivers may not have enough spare capacity to
350 perform even simplified mobile phone tasks. Smartphone applications to support mobile
351 phone integration while driving do not have awareness of driving demands and, therefore,
352 cannot facilitate or inhibit driver decisions to engage in mobile phone tasks or not. This
353 is concerning, as research has demonstrated that driver judgement is not always adequate
354 among some high-risk groups such as novice drivers (Oviedo-Trespalcacios et al., 2018b).

355 Interactions between the external communicators and the smartphone applications
356 are an important influence in the applications' functioning. Smartphone applications
357 frequently interact with external communicators in two ways: to inform them that the
358 driver is unavailable to engage in texting or talking or to redirect the communication
359 activity (e.g., transfer to voice mail). Typically, external communicators will receive a
360 message, often personalised, informing them that the person that they are contacting is
361 driving, and they have the chance to decide whether to still engage in a texting activity
362 (i.e., resend the message) or change the interface (e.g., substitute the text message with a
363 mobile phone call). External communicators with a good understanding of the risks would
364 serve as key actors in moderating driver exposure to mobile phone interactions.

365 Therefore, it is important to broaden educational efforts to educate external
366 communicators of their role in preventing crashes due to mobile phone use while driving.

367 In this research it was identified that smartphone applications have external users
368 who are not directly trying to communicate with the drivers. These external users might
369 include family, friends or employers of the driver. An example of this is the
370 *DriverAlertNow* application that is promoted as having the capability to inform parents
371 of the mobile phone use behaviours of young drivers or whether the application has been
372 overridden/deactivated. Similar performance sharing arrangements are promoted to fleet
373 companies with the intention of providing surveillance of the driving performance of
374 employees (see *Fleetmode* application). Although such applications seem to operate
375 mostly on a punishment model, some promising approaches based on rewards are also
376 emerging among available applications. A good example of this is the *Text ninja*
377 application that is being used for advocacy groups to create collaborative awareness
378 campaigns where a group of drivers can monitor their performance and measure the
379 benefits from using applications to prevent distracting driving. Approaches using
380 incentives have been reported to be effective in improving driver behaviour, for example
381 the Speed Camera Lottery in Stockholm, Sweden (see Schultz (2010)), and various types
382 of Pay As You Drive vehicle insurance schemes (Tselentis et al., 2016). Moreover, incentives
383 may also increase acceptance of technology and could help to support implementation of
384 in-car systems (Vaezipour et al., 2017). This has implications for insurance organizations
385 that could potentially use the information to offer rewards and ultimately save lives.
386 Nonetheless, concerns about privacy could also play an important role in discouraging
387 users (Khalid et al., 2015). Research is necessary to determine schemes of private data
388 management that guarantee driver acceptance while ensuring efficiency in reducing
389 exposure to mobile phone distraction.

390 There is a need for communication between the mobile phone and the vehicle.
391 New vehicles today have the capability to support mobile phone use through vehicle
392 audio systems (also known as integrated hands-free). This is very convenient given that
393 installing external hands-free devices is often a barrier due to cost or lack of practicality.
394 The vehicle could also communicate with the smartphone applications to guarantee a
395 timely activation of the applications just before the vehicle starts moving. Automatic
396 activation of the smartphone application to prevent distracted driving is more reliable than
397 the driver having to launch the application before every drive.

398 The smartphone applications to prevent mobile use while driving identified in this
399 study are voluntary and for personal use. To date, there is little research on drivers'
400 intention to use these smartphone applications to prevent mobile phone use while driving.
401 Previous work has suggested that many of these smartphone applications are difficult to
402 use and not always reliable (Sousa, 2015, Tchankue et al., 2012). In addition, previous
403 efforts using psychosocial variables (i.e., attitudes, likelihood of engaging and frequency
404 of reported texting and driving behaviours) to predict the use of smartphone applications
405 to reduce mobile phone distracted driving have had little success (McGinn, 2014).
406 Another important finding was that nearly half of the applications are free, which might
407 influence drivers' adoption of this technology. Technology acceptance models for in-
408 vehicle technology have highlighted the importance of affordability in the acceptance of

new technologies (Mitsopoulos-Rubens and Regan, 2017). Acceptance of the technology is critical given that many of these applications can be easily overridden or disabled by the driver at any time (e.g. "Do Not Disturb While Driving"), which makes the technology highly dependent on the willingness of the driver to use the application. This highlights the need for a more comprehensive understanding of drivers' requirements and involving them in the design of these smartphone applications to facilitate greater acceptance and adoption of the technology. The importance of user involvement in designing new technologies has been highlighted previously (DIS, 2009, François et al., 2017).

Lastly, it is very encouraging that more stakeholders are systematically collaborating in the design and development of technology to prevent distracted driving. The role of mobile phone manufacturers and application designers in preventing mobile phone distracted driving has tended to be overlooked. Mobile phone companies are seen often as complacent about their responsibility (Galitz, 2018). To date, interventions to stop mobile phone distraction while driving have been heavily focused on the role of the driver, while ignoring the responsibility of the wider road transport or communication authorities (Young and Salmon, 2015, Parnell et al., 2017, Parnell et al., 2016). A systemic approach is more likely to succeed in preventing mobile phone use while driving. This research confirms that the smartphone applications to prevent mobile phone distracted driving can be an important part of a system wide approach involving industry, government, community, and family, given their capabilities to record information and block risky interactions such as visual-manual tasks.

5. Conclusion

This paper reviewed current smartphone applications designed to prevent distracted driving. The results show that the available smartphone applications use blocking approaches rather than manage the workload of mobile phone interactions while driving. This limits the success of the applications given that mobile phone use is becoming more pervasive. Nonetheless, these smartphone applications show the potential of reducing exposure to risky behaviours such as taking one's eyes off the road.

Current approaches to application design show the potential for these applications to interact with aspects of the driver ecosystem such as the vehicle and external communicators and users. This confirms that smartphone applications have the capability to be part of large systemic intervention to prevent mobile phone distraction. However, there are a great number of applications, including some that do not necessarily prevent risky interactions, as demonstrated in this investigation. In order to guarantee a reduction in mobile phone use while driving, a definition of standard practice for the design of these applications is necessary.

One limitation of this study was that user reviews and satisfaction were not considered. In addition, there is no guarantee that the functions mentioned by the applications work appropriately every time. Future research is necessary to measure user perspectives and the usability of the applications as well as the reliability of the smartphone applications.

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602 Appendix A

603 **Table A1.** Web pages for the smartphone applications

Smartphone application name	Web page (Information source)
Android Auto	https://www.android.com/auto/
AT&T DriveMode	https://www.att.com/gen/press-room?pid=23185
BAZZ Text While Driving Safe	http://www.bazz.co/en/
Car Mode	https://play.google.com/store/apps/details?id=com.sec.android.automotive.drivelink&hl=en
Cellcontrol	https://www.cellcontrol.com/
cellMUTE+ Drive Mode & More	https://www.cellmute.com/qa
Do Not Disturb While Driving	https://support.apple.com/en-au/HT208090
Down for the Count	http://downforthecountapp.com/
Drive Now Text Later	https://play.google.com/store/apps/details?id=com.techideas4you.drivenowtextlater&hl=en
Drive Safe	https://play.google.com/store/apps/details?id=westport.andrewirwin.com.drivesafe&hl=en
DriveAlertNow	https://drivealtrnow.com
DriveSafe.ly	http://www.drivesafe.ly/
eBrake	http://ebrake.ca/faq
Fleetmode	https://www.fleetmode.com/how-fleetmode-works
Lifesaver	https://lifesaver-app.com/
Live-n-Drive	https://play.google.com/store/apps/details?id=in.codersunlimited.livendrive
MessageLOUD	http://www.messagecloud.com/android
Mojo: Rewards for Safe Driving	https://itunes.apple.com/us/app/mojo-rewards-for-distraction-free-driving/id1213084199?mt=8
No Texting While Driving App	https://play.google.com/store/apps/details?id=com.driveon
OneTap—Block Phone Alerts	http://www.getonetap.com/
ProtextMe Safe Driving FREE	http://protextme.com/
PULL OVER TO TEXT	http://www.globalmobilealert.com/pull-over-to-text
Safe Driving + Auto SMS	https://play.google.com/store/apps/details?id=com.lemi.safedrivingmsautoreplypro&hl=en
Safe Driving App: Drivemode	https://drivemode.com/
Safely Go	https://play.google.com/store/apps/details?id=com.safely.go.driver.safety.stop.texting.driving&hl=en
Text Ninja/Motivate	https://textninja.com/
TextDrive	https://play.google.com/store/apps/details?id=com.smalltalkapps.textdrive
TextLimit	https://www.textlimit.com/
TrueMotion Family Safe Driving	https://gotruemotion.com/

604

605 **Table A2.** Smartphone applications which were discontinued during this study¹

Smartphone application name	Web page (Information source)
Focus – Screen Free Driving	https://itunes.apple.com/us/app/focus-screen-free-driving/id1033662985?mt=8
Live2Txt	http://www.getlive2txt.com/index.html
Safe Driving - Contacts & Apps	https://play.google.com/store/apps/details?id=com.webillow.safedrive

606

¹ The smartphone applications listed in the table were identified to be discontinued as of the 9th of March 2018 in Australia. Therefore, these applications could not be included in the content analysis.